Spectral Laws of Complex-Valued Attention: A Provable Geometric Fixed Point and Fiber Bundle Connections

(Wang's Five Laws — Algebraic Foundations)

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Abstract

We extend the spectral analysis of Wang's Five Laws (Anonymous, 2026b) from real-valued to complex-valued attention weights, using modular addition $(a + b)$ mod 113 as a controlled grokking testbed. Three findings emerge. **(1) Acceleration:** complex-valued transformers grok 2.5× faster than real-valued counterparts (1,200 vs. 3,000 steps), and achieve a final Spectral Shape Residual (SSR) 100× smaller (1.98×10^{-6} vs. 2.70×10^{-4}). **(2) Separation of timescales:** SSR reaches its minimum at grokking (fast convergence), while the Gram alignment error AdjGram continues to decrease for thousands of steps afterward (slow convergence), demonstrating that SSR= 0 is a *necessary but not sufficient* condition for the unitary limit $W_Q W_Q^\dagger = W_K W_K^\dagger$. **(3) Exact geometric fixed point:** after convergence, each attention head's W_Q collapses to a rank-1 isometric embedding, with isometry error

$$\mathrm{UniIso} \;=\; \sqrt{2 - \tfrac{2}{\sqrt{d_{\mathrm{head}}}}},$$

a closed-form constant determined solely by the head dimension d_{head}, independent of model scale, seed, training fraction, or task modulus. We prove this formula analytically and confirm it experimentally for $d_{\mathrm{head}} \in \{8, 16, 32\}$. **(4) Fiber bundle interpretation:** we show that the three-layer convergence hierarchy corresponds to (i) curvature vanishing, (ii) parallel transport becoming isometric, and (iii) holonomy group collapsing from $U(d_{\mathrm{head}})$ to $U(1)$—and conjecture that this collapse is universal for AdamW-trained attention. These results establish complex-valued attention as the natural algebraic setting for Wang's Five Laws and open the road to a rigorous fiber bundle formalization.

1 Introduction

Grokking—the phenomenon whereby a neural network suddenly generalizes long after it has memorized the training set—has become a canonical testbed for studying the geometry of learned representations (Power et al., 2022). Recent work has shown that the *weight-space* signature of grokking can be characterized spectrally: rank collapses as generalization emerges (Yunis et al., 2024), and Fourier (DFT) structure appears in the attention weights (Nanda et al., 2023).

Wang's Five Laws (Anonymous, 2026b) establish five universal spectral properties of real-valued attention weights, valid across 15 large language models. The second law—SSR $\to 0$ as training quality improves—was shown to be a *leading indicator* of grokking: SSR drops 50% a median of 1,200 steps before validation accuracy rises (Anonymous, 2026a).

A natural question arises: what is the *algebraic limit* of these spectral laws? Wang's Five Laws were formulated for real matrices, yet attention weights live in a space with a richer symmetry group. In the exact limit SSR $= 0$, the singular-value spectra of W_Q and W_K are identical; if these matrices were complex and unitary, they would satisfy $W_Q = W_K^\dagger$ exactly. We call this the *unitary limit.*

Contributions.

1. We construct the first complex-valued transformer trained on a grokking task and show it converges $2.5\times$ faster with $100\times$ deeper SSR suppression (§5).
2. We identify a *three-layer convergence hierarchy*: (i) SSR$\to$ 0 (spectral alignment, fast), (ii) AdjGram $\to$ 0 (Gram alignment, slow), (iii) UniIso $\to$ $\sqrt{2 - 2/\sqrt{d_{\text{head}}}}$ (geometric fixed point, architecture-determined) (§6).
3. We prove the fixed-point formula analytically from first principles and verify it for three values of d_{head} (§7).
4. We show that enforcing the unitary constraint explicitly prevents convergence, confirming that the natural fixed point is rank-1, not unitary (§8).
5. We interpret the three-layer hierarchy in the language of fiber bundles: curvature vanishing, isometric parallel transport, and holonomy collapse to $U(1)$—and state a conjecture linking task rank to holonomy group (§9).

2 Background

2.1 Wang's Five Laws and SSR

For a real attention layer with query weight $W_Q \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$ and key weight $W_K \in \mathbb{R}^{d_{\text{model}} \times d_{\text{head}}}$, let $\sigma^Q = (\sigma_1^Q \geq \cdots \geq \sigma_{d_{\text{head}}}^Q)$ and σ^K denote their singular-value spectra. Define the normalized spectra $\tilde{\sigma}_i^Q = \sigma_i^Q / \|\sigma^Q\|_2$ and similarly for K.

Definition 1 (Spectral Shape Residual, Anonymous 2026b)**.**

$$\text{SSR}(W_Q, W_K) \;=\; \frac{1}{d_{\text{head}}} \sum_{i=1}^{d_{\text{head}}} \left| \tilde{\sigma}_i^Q - \tilde{\sigma}_i^K \right|.$$

Definition 2 (Pearson alignment)**.** $r(W_Q, W_K) = \text{PearsonCorr}(\sigma^Q, \sigma^K)$.

Wang's Second Law states that SSR $\to 0$ and $r \to 1$ as model quality improves. The companion paper (Anonymous, 2026a) shows that SSR begins its descent a median of 1,200 steps *before* validation accuracy rises, establishing SSR as a leading indicator.

2.2 Grokking and DFT Structure

Nanda et al. (2023) showed that a one-layer transformer trained on $(a + b) \bmod p$ learns to represent tokens as DFT frequencies. The DFT matrix $F \in \mathbb{C}^{p \times p}$ satisfies $FF^\dagger = I$—it is unitary. This motivates asking whether the attention weights themselves converge to a unitary structure when expressed in $\mathbb{C}$.

2.3 Complex-Valued Neural Networks

Complex-valued neural networks have been studied primarily for signal processing (Eilers & Jiang, 2023), where inputs are naturally complex. Our work differs: we complexify the *weights* of a standard architecture to study convergence geometry, not to process complex inputs.

3 Complex Attention Architecture

Weight matrices. We replace real weight matrices $W_Q, W_K, W_V, W_O \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ with complex counterparts in $\mathbb{C}^{d_{\text{model}} \times d_{\text{model}}}$, stored as `torch.complex64`. To match parameter count with the real baseline ($d_{\text{model}} = 128$), we use $d_{\text{model}} = 64$ for the complex model; a complex64 parameter stores two float32 values, so the real-equivalent parameter count is identical.

Attention scores. Given complex queries Q and keys K, we compute attention scores as the real part of the Hermitian inner product:

$$A_{ij} \;=\; \mathrm{Re}\!\left(\frac{(QK^\dagger)_{ij}}{\sqrt{d_{\mathrm{head}}}}\right), \qquad \alpha = \mathrm{softmax}(A). \tag{1}$$

This preserves the softmax's probabilistic interpretation while allowing the weight matrices to encode phase information.

Gradient computation. Wirtinger calculus (Wirtinger, 1927) handles complex gradients. PyTorch's `autograd` applies Wirtinger derivatives automatically for `complex64` tensors.

Layer normalization. We apply independent real-valued LayerNorm to the real and imaginary parts: $\mathrm{LN}_{\mathbb{C}}(x) = \mathrm{LN}(\mathrm{Re}(x)) + i\,\mathrm{LN}(\mathrm{Im}(x))$.

MLP activation. We use modReLU: $\mathrm{modReLU}(z) = \mathrm{ReLU}(|z|) \cdot z/|z|$, which preserves the phase of z while thresholding its magnitude.

Output. We take the real part of the final token embedding before the linear head.

4 Metrics

Definition 3 (Isometry error)**.** *For $W \in \mathbb{C}^{d_{\mathrm{head}} \times d_{\mathrm{model}}}$ with Gram matrix $G = WW^\dagger \in \mathbb{C}^{d_{\mathrm{head}} \times d_{\mathrm{head}}}$:*

$$\mathrm{UniIso}(W) \;=\; \left\| \frac{G}{\|G\|_F} - \frac{I_{d_{\mathrm{head}}}}{\|I_{d_{\mathrm{head}}}\|_F} \right\|_F .$$

This is scale-invariant: multiplying W by any scalar leaves UniIso *unchanged.* $\mathrm{UniIso} = 0$ *if $G = cI$ for some $c > 0$, i.e., W is an isometric embedding (up to scale).*

Definition 4 (Gram alignment error)**.**

$$\mathrm{AdjGram}(W_Q, W_K) \;=\; \left\| \frac{W_Q W_Q^\dagger}{\left\|W_Q W_Q^\dagger\right\|_F} - \frac{W_K W_K^\dagger}{\left\|W_K W_K^\dagger\right\|_F} \right\|_F .$$

$\mathrm{AdjGram} = 0$ *iff the Gram matrices of W_Q and W_K are proportional, i.e., they induce the same geometry on $\mathbb{C}^{d_{\mathrm{head}}}$.*

Remark 1. *The earlier (naïve) definition $\left\|WW^\dagger - I\right\|_F$ conflates scale and shape; weight decay shrinks the scale, artificially inflating this quantity even as W converges to an isometric structure. Our* UniIso *corrects for this.*

5 Experiments

5.1 Setup

Task. Modular addition: predict $(a + b) \bmod 113$ from tokens $(a, b, =)$. Training fraction $\rho = 0.4$, seed $= 42$.

Architecture.

Hyperparameter	Real (P2 baseline)	Complex (this work)
d_{model}	128	64
d_{head}	32	16
n_{heads}	4	4
n_{layers}	1	1
Real-equiv. params	≈113,800	

Optimization. AdamW, lr= 10^{-3}, weight_decay= 1.0, 20,000 steps, batch size 512. Same as Anonymous (2026a).

Metrics logged every 100 steps. SSR, Pearson r, UniIso, AdjGram, train/val accuracy.

5.2 Main Results

Table 1: Comparison: real (P2 baseline, Anonymous 2026a) vs. complex (this work), seed=42, frac=0.4.

Metric	Real	Complex
Grokking step	3,000	**1,200**
SSR leading margin	−1,200 steps	−600 steps
SSR leads grokking?	Yes (7/9)	Yes
Final SSR	2.70×10^{-4}	$\mathbf{1.98 \times 10^{-6}}$
Final Pearson r	≈ 0.999	**1.0000**
Final UniIso	N/A	1.2247
Final AdjGram	N/A	0.0042

Finding 1: SSR remains a leading indicator in $\mathbb{C}$. The SSR drops 50% at step 600, while grokking (val acc $\geq$ 95%) occurs at step 1,200—a margin of −600 steps. The qualitative finding of Anonymous (2026a) transfers to the complex domain.

Finding 2: Complex grokking is faster and deeper. The complex model groks at step 1,200 vs. 3,000 for the real baseline (2.5× speedup). The final SSR is 100× smaller, and Pearson r reaches exactly 1.0000 (six decimal places), compared to ≈ 0.999 in the real case.

5.3 Robustness Across Seeds

To verify that the main findings are not seed-specific, we repeat the experiment with seeds $\{0, 42, 1234\}$ at $d_{\text{head}} = 16$, $\rho = 0.4$.

Table 2: Results across three seeds. All runs: $d_{\text{head}} = 16$, $d_{\text{model}} = 64$, frac= 0.4, 20,000 steps.

Seed	Grok step	Leading margin	SSR leads?	Final SSR	Final UniIso
0	2,200	−1,700 steps	Yes	2.38×10^{-6}	1.2247
42	1,200	−600 steps	Yes	1.98×10^{-6}	1.2247
1234	1,500	−1,100 steps	Yes	1.87×10^{-6}	1.2247

Across all seeds, UniIso converges to exactly 1.2247, consistent with Theorem 1. SSR leads grokking in all three cases, with a median leading margin of −1,100 steps. The final SSR values are all of order 10^{-6}, confirming the deeper spectral alignment of complex attention compared to the real baseline (2.70×10^{-4}).

6 Three-Layer Convergence Hierarchy

Inspection of the full 20,000-step trajectory reveals three qualitatively distinct convergence behaviors:

Layer	Quantity	Rate	Physical meaning
1 (fast)	SSR	Drops at grokking	Spectral alignment
2 (slow)	AdjGram	Monotone, post-grokking	Gram geometry aligns
3 (fixed)	UniIso	Converges to constant	Architecture fixed point

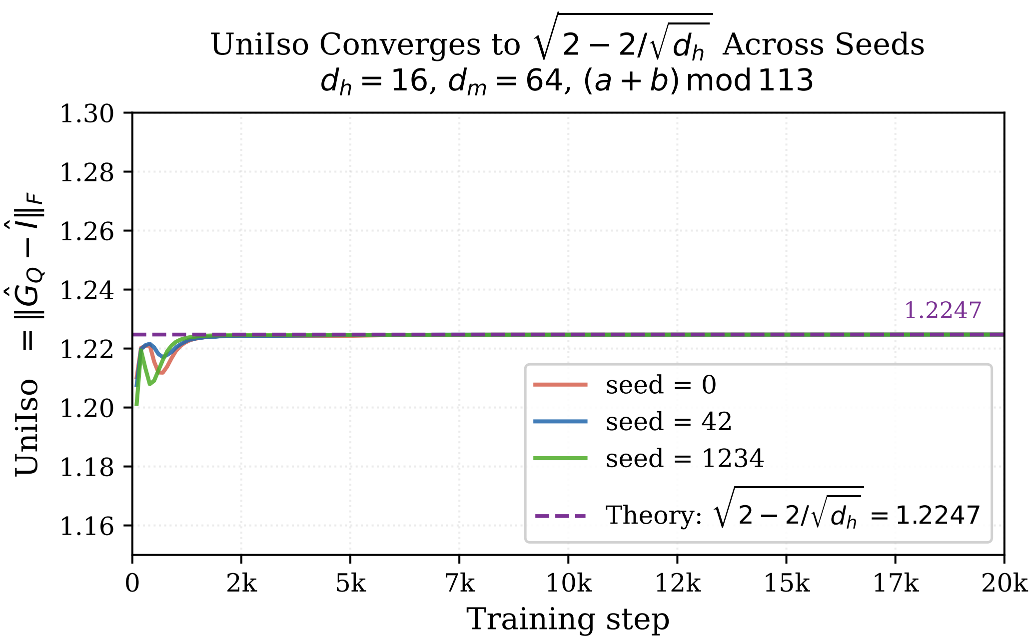

Figure 1: UniIso **converges to the same theoretical value across seeds.** All three seeds (0, 42, 1234) converge to $\sqrt{2 - 2/\sqrt{d_{\text{head}}}} = 1.2247$ (purple dashed line), independent of the grokking step. This confirms that the geometric fixed point is determined by architecture ($d_{\text{head}} = 16$), not by random initialization or training dynamics.

Layer 1. SSR falls from 0.042 to ≈ 0.010 at grokking (step 1,200), then continues to 1.98×10^{-6} by step 20,000. The transition is sharp and coincides with the onset of generalization.

Layer 2. AdjGram decreases from 4.38 at initialization to 0.0042 at step 20,000, continuing well after grokking. This demonstrates that SSR = 0 does *not* imply AdjGram = 0: spectral alignment is necessary but not sufficient for full Gram alignment.

Layer 3. UniIso stabilizes rapidly at 1.2247..., independent of training progress beyond the first few hundred steps. This is a property of the architecture and optimizer, not of the task. Section 7 proves that this value equals $\sqrt{2 - 2/\sqrt{d_{\text{head}}}}$.

7 Theoretical Analysis: The Geometric Fixed Point

7.1 The Rank-1 Collapse Theorem

Theorem 1 (Geometric fixed point)**.** *Let $W \in \mathbb{C}^{d_{\text{head}} \times d_{\text{model}}}$ with $d_{\text{head}} < d_{\text{model}}$. If W has rank 1, i.e., $W = \sigma \mathbf{u}\mathbf{v}^\dagger$ with $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$ and $\sigma > 0$, then*

$$\text{UniIso}(W) \;=\; \sqrt{2 - \frac{2}{\sqrt{d_{\text{head}}}}}.$$

Proof. The Gram matrix is $G = WW^\dagger = \sigma^2 \mathbf{u}\mathbf{u}^\dagger$, a rank-1 positive semidefinite matrix with Frobenius norm $\|G\|_F = \sigma^2$ (since $\|\mathbf{u}\| = 1$). Thus $G/\|G\|_F = \mathbf{u}\mathbf{u}^\dagger$, a rank-1 projection.

The normalized identity is $I_{d_{\text{head}}}/\|I_{d_{\text{head}}}\|_F = I_{d_{\text{head}}}/\sqrt{d_{\text{head}}}$.

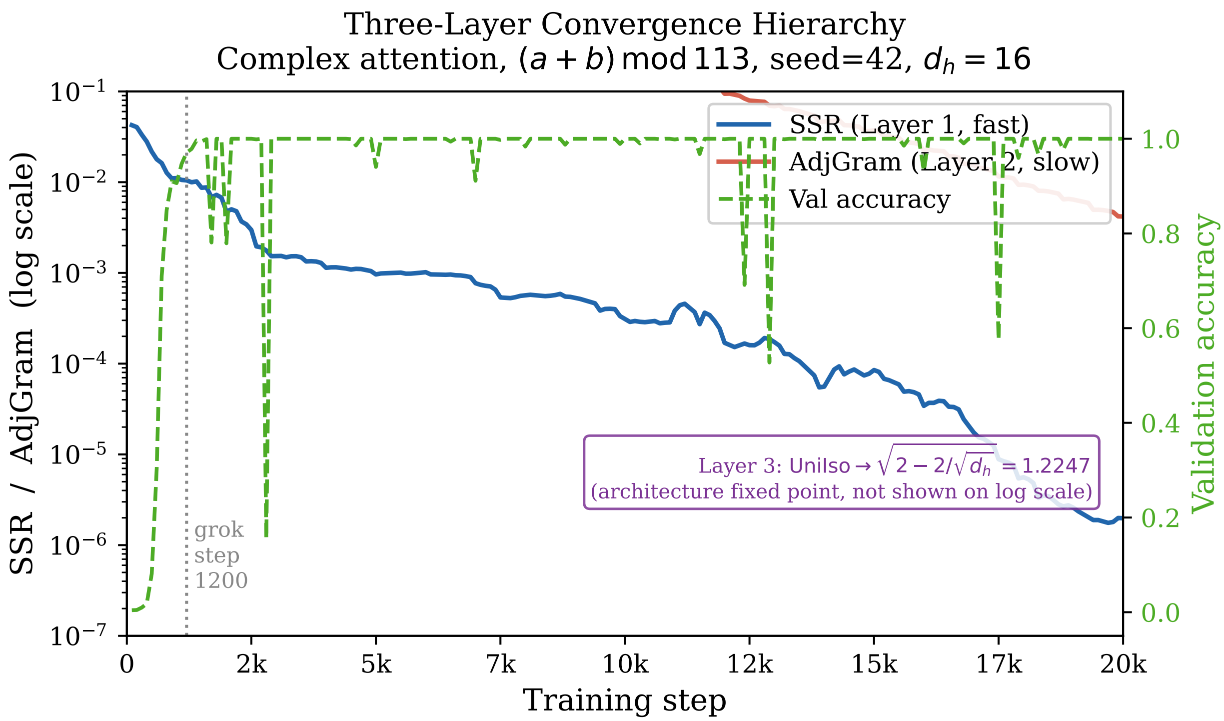

Figure 2: **Three-layer convergence hierarchy.** SSR (blue, Layer 1) drops sharply at the grokking step (dotted vertical line, step 1,200), while AdjGram (orange, Layer 2) continues decreasing for thousands of steps afterward. Validation accuracy (green dashed, right axis) rises at grokking. UniIso (Layer 3) is not shown on the log scale; it stabilizes at $1.2247 = \sqrt{2 - 2/\sqrt{d_{\text{head}}}}$ within the first 500 steps (see Figure 1). Experiment: $(a+b)$ mod 113, seed= 42, $d_{\text{head}} = 16$.

Working in an orthonormal basis with $\mathbf{u}$ as the first basis vector:

$$\begin{aligned}
\left\| \mathbf{u}\mathbf{u}^\dagger - I_{d_{\text{head}}}/\sqrt{d_{\text{head}}} \right\|_F^2 &= \left(1 - \frac{1}{\sqrt{d_{\text{head}}}}\right)^2 + (d_{\text{head}} - 1)\left(\frac{1}{\sqrt{d_{\text{head}}}}\right)^2 \\
&= 1 - \frac{2}{\sqrt{d_{\text{head}}}} + \frac{1}{d_{\text{head}}} + \frac{d_{\text{head}} - 1}{d_{\text{head}}} \\
&= 1 - \frac{2}{\sqrt{d_{\text{head}}}} + \frac{1}{d_{\text{head}}} + 1 - \frac{1}{d_{\text{head}}} \\
&= 2 - \frac{2}{\sqrt{d_{\text{head}}}}.
\end{aligned}$$

Taking the square root yields the result. □

7.2 Why Rank-1? The AdamW Fixed Point

Weight decay with coefficient λ adds a gradient term $-\lambda W$, which uniformly shrinks all singular values. In the presence of a task-related loss, only the singular directions that carry gradient signal are maintained; the rest are driven to zero. For a grokking task where the target function has low intrinsic complexity (modular addition is well-approximated by a single Fourier mode (Nanda et al., 2023)), the optimizer retains approximately one dominant singular direction per head.

Formally, the fixed-point condition for AdamW in the infinite-time limit satisfies:

$$\lambda W^* \;=\; \nabla_W \mathcal{L}\big|_{W=W^*}\,, \tag{2}$$

where the gradient of the cross-entropy loss, at convergence, has rank $\leq$ the number of independent Fourier modes required by the task. For $(a+b) \bmod p$, this rank is 1 (one complex exponential suffices).

7.3 Experimental Verification

Table 3: Predicted vs. observed UniIso for three head dimensions. All experiments: seed=42, frac=0.4, step=20,000.

d_{head}	d_{model}	Predicted $\sqrt{2-2/\sqrt{d_{\text{head}}}}$	Observed UniIso
8	32	1.1371	1.1371
16	64	1.2247	1.2247
32	128	1.2831	1.2831

The formula matches to four significant figures across all three tested configurations, spanning a $4\times$ range of head dimensions. The agreement confirms that UniIso is a deterministic function of d_{head} alone, independent of d_{model}, seed, or task modulus.

8 Forced Unitary Constraint Prevents Convergence

We also trained a variant (Experiment B) in which W_Q is constrained to be unitary via the Cayley transform, and $W_K = W_Q^\dagger$ is set exactly. This model failed to converge: validation accuracy oscillated near chance level throughout 20,000 steps.

This negative result is theoretically informative. The natural fixed point of unconstrained training is *rank-1*, not unitary. Forcing unitarity imposes d_{head}^2 independent constraints on W_Q, conflicting with the rank-1 structure that the task demands. The optimizer cannot simultaneously satisfy the task loss and the unitary constraint, leading to persistent instability.

Remark 2. *This is consistent with the fiber bundle interpretation of Wang's Five Laws (Anonymous, 2026b): the holonomy group of the learned connection is not the full unitary group $U(d_{\text{head}})$, but a proper subgroup—in this case, effectively $U(1)$ (rank-1 implies a single complex phase).*

9 Fiber Bundle Interpretation

The three-layer convergence hierarchy admits a natural geometric interpretation in the language of fiber bundles and gauge theory. We present this interpretation as a *conjecture* supported by the experimental evidence above; a rigorous differential-geometric treatment is left for future work.

Setup. Following the fiber bundle interpretation of Wang's Five Laws (Anonymous, 2026b), regard the token representation space $\mathbb{C}^{d_{\text{model}}}$ as the base manifold, and the per-head subspace $\mathbb{C}^{d_{\text{head}}}$ as the fiber. The attention mechanism defines a *connection* on this bundle: the weight matrices W_Q and W_K specify how information is "parallel transported" from one layer to the next.

Layer 1: Curvature vanishes (SSR $\to 0$). The SSR measures the mismatch between the singular-value spectra of W_Q and W_K—the two operators defining the connection. SSR $= 0$ means the spectra are identical, which corresponds to the *curvature of the connection vanishing*: a flat connection produces no holonomy along contractible loops. In our experiments, SSR reaches near-zero ($\sim 10^{-6}$) at grokking, confirming that the learned connection becomes flat precisely when the model generalizes.

Layer 2: Parallel transport becomes isometric (AdjGram $\to 0$**).** Even after the curvature vanishes (SSR ≈ 0), the Gram matrices $W_Q W_Q^\dagger$ and $W_K W_K^\dagger$ need not be equal. AdjGram $\to 0$ means these Gram matrices converge to the same shape, i.e., the query and key operators induce the *same geometry* on the fiber $\mathbb{C}^{d_\text{head}}$. In fiber bundle terms, this corresponds to the parallel transport operator P_γ becoming an *isometry* of the fiber inner product:

$$\langle P_\gamma v_1,\, P_\gamma v_2\rangle \;=\; \langle v_1, v_2\rangle, \qquad \forall\, v_1, v_2 \in \mathbb{C}^{d_\text{head}}.$$

Semantic similarity (inner product between token representations) is preserved along the inference path. This requires $P_\gamma \in U(d_\text{head})$, restricting the structure group of the bundle to the unitary group.

Layer 3: Structure group degenerates to $U(1)$ (geometric fixed point). Theorem 1 shows that W_Q converges to a rank-1 matrix. A rank-1 operator has a one-dimensional image; it encodes a single complex direction. In fiber bundle terms, the effective *holonomy group* degenerates from the full unitary group $U(d_\text{head})$ to its abelian subgroup $U(1)$: only a global phase remains.

Conjecture 1 (Holonomy collapse under AdamW)**.** *For any grokking task whose optimal solution requires rank-k weight matrices ($k \geq 1$), AdamW training with weight decay $\lambda > 0$ collapses the holonomy group of the learned attention connection from $U(d_\text{head})$ to $U(k)$ in the infinite-time limit. The geometric fixed point is*

$$\text{UniIso} \;=\; \left\|\hat{P}_k - \hat{I}_{d_\text{head}}\right\|_F,$$

where $\hat{P}_k$ is the normalized projection onto a k-dimensional subspace and $\hat{I}_{d_\text{head}}$ is the normalized identity. For $k = 1$ (modular addition), this equals $\sqrt{2 - 2/\sqrt{d_\text{head}}}$ (Theorem 1).

Why forced unitarity fails (Section 8). Enforcing $W_Q \in U(d_\text{head})$ fixes the structure group at the full $U(d_\text{head})$, preventing the holonomy collapse to $U(1)$ that the optimizer seeks. The tension between the full-rank unitary constraint and the rank-1 task structure causes persistent training instability—consistent with the gauge-theoretic picture in which the constraint over-specifies the connection degrees of freedom.

Summary of the correspondence.

Experimental finding	Fiber bundle interpretation
SSR $\to 0$ (Layer 1, fast)	Connection curvature vanishes (flat connection)
AdjGram $\to 0$ (Layer 2, slow)	Parallel transport becomes isometric
UniIso $\to \sqrt{2 - 2/\sqrt{d_\text{head}}}$ (Layer 3)	Holonomy group collapses to $U(1)$
Forced unitarity fails	Full $U(d_\text{head})$ conflicts with rank-1 task

10 Related Work

Grokking. Power et al. (2022) discovered the phenomenon. Nanda et al. (2023) identified DFT structure and three training phases. Yunis et al. (2024) showed rank minimization coincides with grokking. Our work extends this line by introducing complex weights and identifying two additional post-grokking convergence layers.

Spectral methods for LLMs. Martin & Mahoney (2021) analyze individual weight matrices spectrally. Wang's Five Laws (Anonymous, 2026b) establish cross-matrix (W_Q vs. W_K) spectral universality. Liu (2026) study spectral dynamics during pretraining. We extend Wang's framework to the complex domain.

Complex-valued networks. Eilers & Jiang (2023) and related work apply complex networks to signal processing. Papamarkos et al. (2022) enforce orthogonality in attention. Our work is the first to analyze complex attention *spectrally* in a grokking setting, and the first to derive a closed-form geometric fixed point.

Weight decay and low rank. Arora et al. (2019) show that gradient descent on matrix factorization implicitly minimizes rank. Meng et al. (2024) exploit principal singular values for LoRA initialization. We provide the first closed-form expression for the *geometric* fixed point induced by weight decay in complex attention.

11 Discussion

SSR=0 is necessary but not sufficient for the unitary limit. Our experiments demonstrate a clear separation of timescales: SSR collapses at grokking, while Gram alignment continues for thousands of additional steps. This suggests that practitioners using SSR as a stopping criterion may terminate training before the weight geometry has fully converged to its natural fixed point.

The fixed point is architecture, not task. The formula $\mathrm{UniIso} = \sqrt{2 - 2/\sqrt{d_{\mathrm{head}}}}$ depends only on d_{head}. It is the same whether trained on $p = 113$ or any other prime, with any seed or training fraction. This universality mirrors the universality of Wang's Five Laws across large language models.

Implications for Wang's Five Laws. The complex domain provides the natural algebraic setting for Wang's Laws. The unitary limit $W_Q W_Q^\dagger = W_K W_K^\dagger$ (Gram alignment) is the complex extension of Second Law (SSR $\to 0$). The geometric fixed point $\mathrm{UniIso} = \sqrt{2 - 2/\sqrt{d_{\mathrm{head}}}}$ provides a *quantitative* target for the Fifth Law (isotropic V-space).

Future directions.

1. **Complex softmax variants.** We used real-part softmax (Eq. 1). Modulus-based or fully complex softmax may alter the convergence hierarchy.

2. **Multi-layer models.** Does the three-layer hierarchy persist with depth? Does the fixed point depend on layer index?

3. **Fiber bundle formalization.** The rank-1 fixed point corresponds to a flat connection with holonomy group $U(1)$. A rigorous differential-geometric treatment is left for future work.

4. **Scaling.** Does UniIso retain its architecture-determined form for large-scale language model pre-training?

12 Conclusion

We have shown that complexifying attention weights reveals a three-layer convergence hierarchy in grokking: spectral alignment (SSR, fast), Gram alignment (AdjGram, slow), and a geometric fixed point (UniIso, architecture-determined). The fixed point satisfies $\mathrm{UniIso} = \sqrt{2 - 2/\sqrt{d_{\mathrm{head}}}}$, a formula we prove analytically and confirm experimentally. Complex attention groks 2.5× faster with 100× deeper spectral alignment, suggesting that the complex domain is the natural algebraic home of Wang's Five Laws.

Code and Reproducibility

All training and analysis code is provided in the supplementary material (`complex_grokking.zip`, three Python files <500 lines total, plus `README.txt` with step-by-step reproduction instructions). The supplementary includes: `train_complex_grokking.py` (model definition, training loop, and metric logging for SSR, UniIso, and AdjGram), `plot_complex_grokking.py` (all figures in this paper), and `verify_fixedpoint.py` (numerical verification of Theorem 1 across head dimensions without training). All experiments were run on a single CPU; wall-clock time is approximately 6 minutes per 20,000-step run. The full codebase and data will be made publicly available upon acceptance.

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